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CONNECT4 PROJECT

ERAU - Prescott | CS225

Presented by Cooper Nelson and William Tinker

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ABOUT US

Cooper Nelson

Major: Aerospace Engineering

Minor(s): Computer Science and Robotics

Year: Sophomore



William Tinker

Major: Mechanical Engineering

Minor(s): Computer Science

Year: Sophomore



PROJECT INTRODUCTION

Connect4 is a classic childhood board game you likely have played with friends and family where your main goal is to connect four of your pieces, vertically, horizontally, or diagonally, before the other player.



PROJECT GOALS

1.
Create a playable connect4 game with C++ using all of the topics we have discussed this semester in CS225
2.
Create a computer bot that you can play against that makes informed decisions based on your moves and its own, rather than placing pieces randomly
3.
Make the bot actually play well enough to beat human players

PROJECT TIMELINE

Week 1:

- Brain storm project ideas:

Choose Notes

Week 2:

- Change project to Connect4
- Outline Project goals

Week 3:

- Begin developing files
- Create front end and begin mapping bot

Week 4:

- Spring Break
- Research bot weighting method

Week 5:

- Get connect4 working in its entirety(even if not perfect)

Week 6:

- Finish full debugging and finalize code
- Create Presentation

ABOUT THE BOT

Class Structure

- All of the checks and data are members of the board class
- The Bot class has two additional Board objects for move simulation
- A separate BotTester class exists for testing & troubleshooting the Bot's behavior
- We use a weight-based approach to determine the optimal move for the bot.



ABOUT THE BOT

calculateWeights

```
try {
    tempBoard2.dropPiece(j, 1);
    if (tempBoard2.checkPlayerWin()) {
        weights[i] -= 100000;
        continue;
    }
    weights[i] += (1 * tempBoard2.checkAllConnections(2) + 1 * (botHighest - tempBoard2.checkHighestConnection(2)));
    weights[i] -= (1 * tempBoard2.checkAllConnections(1) + 3 * (playerHighest - tempBoard2.checkHighestConnection(1)));
}
catch (std::out_of_range& e) {
    weights[i] += (1 * tempBoard2.checkAllConnections(2) + 1 * (botHighest - tempBoard2.checkHighestConnection(2)));
    weights[i] -= (1 * tempBoard2.checkAllConnections(1) + 3 * (playerHighest - tempBoard2.checkHighestConnection(1)));
    continue;
}
}

weights[1] += 1;
weights[2] += 2;
weights[3] += 3;
weights[4] += 2;
weights[5] += 1;
```

Latest version of calculate weights

```
for (int i = 0; i < COLUMNS; i++) {
    weights[i] = 0;
    tempBoard = trueBoard;
    //drop a bot piece in column i, if bot win, add 10000 weight
    tempBoard.drop_piece(i, 2);
    if (tempBoard.check_bot_win()) {
        weights[i] += 10000;
    }
    for (int j = 0; j < COLUMNS; j++) {
        //drop a player piece in column j
        // if player win, subtract 1000 weight
        //add all bot connections to weight
        //subtract all player connections from weight
        tempBoard2 = tempBoard;
        tempBoard2.drop_piece(j, 1);
        if (tempBoard2.check_player_win()) {
            weights[i] -= 1000;
        }
        weights[i] += tempBoard2.check_all_connections(2);
        weights[i] -= tempBoard2.check_all_connections(1);
    }
}
```

Initial Version

LIVE DEMO



NOTABLE CHALLENGES

1. File I/O

Figuring out the best way to write all of the information we wanted to the file, in a way that created a log required using **IOS::app** to properly append and create a log

2.

Opening Bot moves were very bad initially

```
Weights: -26 -26 -10 -12 -8 -24 -26
1 | * * * * * * *
2 | * * * * * * *
3 | * * * * * * *
4 | * * * * * * *
5 | * * * X * * *
6 | * * * 0 0 * *
+-----+
  1 2 3 4 5 6 7
```

```
Weights: -17 -16 -6 -7 -7 -14 -17
1 | * * * * * * *
2 | * * * * * * *
3 | * * * * * * *
4 | * * * * * * *
5 | * * * X * * *
6 | * * * 0 0 * *
+-----+
  1 2 3 4 5 6 7
```

CONCLUSION

Overall take away:

- C++ is much easier than C
- Our bot performed beyond our hopes
- Overloading is still an iffy topic but is incredibly helpful
- We are not good at connect4...

THANK YOU

Any Questions?

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